

Occurrence, virulence, and AMR profile of *Vibrio parahaemolyticus* isolated from shellfish growing areas located along the southwest coast of India

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ABSTRACT

Vibrio parahaemolyticus is a leading cause of human gastroenteritis associated with seafood consumption. The present study aimed to investigate the occurrence and risk assessment of *V. parahaemolyticus* isolated from live Indian black clams, sediment, and water samples collected from shellfish harvesting areas located along the southwest coast of India. Out of the total 72 samples collected, 55.6% revealed the presence of *V. parahaemolyticus*; the highest occurrence was observed in shellfish samples. The presence of *tdh* and *trh* virulence genes was screened by multiplex PCR. Virulence genes could be detected in 25.8% of the strains; 19.35% of them were *trh* positive and 3.2% were *tdh* positive, while 3.2% of strains exhibited the coexistence of both virulence genes. Antimicrobial resistance (AMR) of the isolates was determined by the disk diffusion method. About 87% of the strains were multiple drug resistant and exhibited 21 diverse antibiotic resistance patterns. The overall multiple antibiotic resistance (MAR) index values ranged from 0 to 0.8. To the best of our knowledge, this is the first report to document the presence of pathogenic and multidrug-resistant *V. parahaemolyticus* in shellfish harvesting areas of the Indian sub-continent. The study reveals possible health hazards associated with consuming shellfish harvested from the study area.

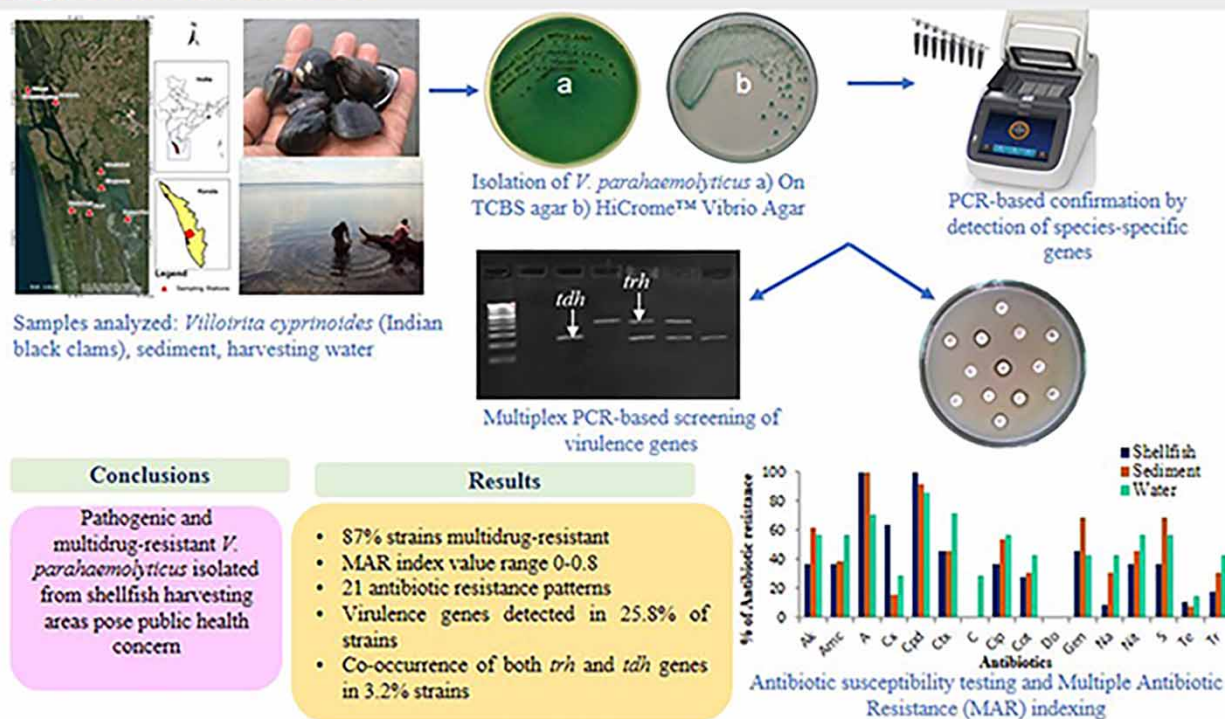
Key words: antimicrobial resistance, bivalve shellfish, estuary, MAR index, *Vibrio parahaemolyticus*, virulence

HIGHLIGHTS

- Pathogenic *V. parahaemolyticus* isolated from shellfish harvesting areas of the Cochin estuary.
- Chromogenic agar, PCR detection of 16S rRNA, *tlh*, and *toxR* genes used for the detection.
- Multiplex PCR revealed *tdh*, *trh* genes, and coexistence of virulence genes.
- 87% of strains were MDR, 21 antibiotic resistance patterns, and high MAR index values were observed.
- The first report of pathogenic *V. parahaemolyticus* from shellfish harvesting areas of India.

GRAPHICAL ABSTRACT

Occurrence, virulence and AMR profile of *Vibrio parahaemolyticus* isolated from shellfish growing areas located along the south-west coast of India



1. INTRODUCTION

Vibrio parahaemolyticus is a Gram-negative halophilic bacterium, widely disseminated in estuarine, marine, and coastal surroundings (Ceccarelli *et al.* 2013; Letchumanan *et al.* 2014). It has been identified as one of the prime causes of human gastroenteritis associated with seafood consumption worldwide (Odeyemi & Stratev 2016). Though foodborne outbreaks associated with *V. parahaemolyticus* have been reported in many Asian countries including India, China, Japan, and Bangladesh (Narayanan *et al.* 2020; Siddique *et al.* 2021), the frequency of detection is low and the number of illnesses tends to be underreported (Scallan *et al.* 2011). Shellfish, being filter feeders, tend to accumulate various contaminants and pathogenic microbes, including vibrios, from the surrounding waters (Pavoni *et al.* 2013). Therefore, consuming raw or undercooked shellfish has been identified as the primary cause of numerous foodborne disease outbreaks (DePaola *et al.* 2000). Being a part of the indigenous microflora of coastal waters, the presence of *V. parahaemolyticus* in shellfish harvesting areas is quite natural, however, their increasing incidence, as well as the presence of virulence factors and multidrug resistance, are an indication of the probable food safety hazards posed to the aquaculture industry and consumers (Siddique *et al.* 2021).

Seasonal influence has been observed in the prevalence of *V. parahaemolyticus* in shellfish and coastal waters; higher bacterial counts were observed at higher surface water temperatures (Urquhart *et al.* 2015). Increased detection of *V. parahaemolyticus* was observed in shellfish, sediment, and harvesting waters during warmer temperatures (Cantet *et al.* 2013).

Pathogenic *V. parahaemolyticus* strains have been reported from clinical as well as environmental sources worldwide (Alam *et al.* 2017). The pathogenic *V. parahaemolyticus* is characterized by several virulence factors such as ToxR, type III secretory system, and Type VI secretion system. However, virulence-associated genes such as the thermostable direct haemolysin gene (*tdh*) and its homolog, the thermostable direct haemolysin-related haemolysin gene (*trh*), are considered the major markers to detect pathogenic *V. parahaemolyticus* (Tada *et al.* 1992). The *tdh* and *trh* genes code for porin-like channel-forming proteins that aggregate and insert into cell membranes, leading to an influx of water and nonspecific efflux of

divalent cations resulting in host cell disruption (Yanagihara *et al.* 2010). However, the majority of the environmental strains are found to be nontoxigenic, lacking virulence genes, while their prevalence in clinical sources is reported relatively more (DePaola *et al.* 2000; Ceccarelli *et al.* 2013). Nevertheless, nontoxigenic *V. parahaemolyticus* strains lacking these toxin-producing genes, yet cytotoxic and capable of causing acute gastroenteritis have been reported (Ottaviani *et al.* 2012). The increasing emergence of multidrug-resistant (MDR) strains among *V. parahaemolyticus* in food and environmental sources is a matter of serious concern (Letchumanan *et al.* 2014; Narayanan *et al.* 2020).

Cochin estuary is one of the largest estuaries located on the southwest coast of India and is well-known for its vast biodiversity. It is a rich natural resource of shellfish and finfish on which the local population depend for their livelihood. The area is of vital importance to the economy of the state of Kerala as the majority of aquaculture products from the area are exported worldwide. There is an increasing number of consignment detention and rejection cases from Asia due to the presence of *V. parahaemolyticus* (Sujeewa *et al.* 2009). Previous studies have reported the presence of drug-resistant vibrios and other bacteria from retail seafood markets in the study area (Sudha *et al.* 2014; Silvester *et al.* 2015). Currently, there is limited information available regarding the occurrence and characteristics of pathogenic *V. parahaemolyticus* in the shellfish harvesting environments from the Indian subcontinent to date.

In this context, a year-long study was conducted to investigate the presence of pathogenic and MDR *V. parahaemolyticus* in the harvesting waters, sediments, and live shellfish samples collected during various seasons, from the harvesting areas of the Cochin estuary. The isolates were screened for the presence of virulence genes and multidrug resistance to understand the potential food safety hazards posed if any, by the consumption of shellfish harvested from the study area.

2. MATERIALS AND METHODS

2.1. Study area and sample collection

Live Indian black clams (*Villorita cyprinoides*), sediment, and harvesting water samples collected from eight selected shellfish harvesting areas located along the Cochin estuary were analysed for 1 year (February 2013–January 2014). The sampling areas included Aroor (1), Valanthakad (2), Cheppanam (3), Valappu (4), South Malipuram (5), Kumbalangi (6), South Vaduthala (7), and Perumpalam (8) as shown in Figure 1. Season-wise study was conducted, namely, pre-monsoon (February to May), monsoon (June to September), and post-monsoon (October to January). One sample of shellfish, sediment, and harvesting waters was collected from every area during these three seasons. A total of 72 samples were analysed, which comprised shellfish ($n = 24$), sediment ($n = 24$), and harvesting water ($n = 24$) samples. Shellfish and sediment samples were collected in sterile polythene bags and harvested water samples in sterile plastic bottles (1,000 mL). All the samples were transported to the laboratory in an ice box and the bacteriological analysis was conducted within 2 h of collection.

2.2. Isolation of *V. parahaemolyticus* from shellfish, sediment, and harvesting waters

Bacteriological analysis was done immediately upon the arrival of samples in the laboratory. Pre-enrichment of all the samples was done in alkaline peptone water (APW) with 3% NaCl at 37 °C for 24 h as described below. Twenty-five grams of aseptically shucked shellfish meat and liquor was transferred to a sterile polythene stomacher bag and homogenized in a stomacher (IUL Instruments, Spain) with 225 mL sterile APW for 1 min, followed by incubation at 37 °C for 24 h. Pre-enrichment of sediment samples was performed by adding 10 g of the collected sediment to 90 mL sterile APW. Approximately 1 L of water samples were filtered through sterile 0.45 µm membrane filters, which were then cut into pieces and pre-enriched in 100 mL of APW. Inoculated APW broths were incubated at 37 °C for 24 h and then a loopful from each was streaked onto Thiosulfate Citrate Bile Salts Sucrose (TCBS) agar (HiMedia, India) and incubated at 37 °C for 24–48 h. Characteristic five sucrose-negative green colonies were picked from each TCBS agar plate and were subjected to Gram staining, followed by preliminary biochemical screening based on oxidase test, catalase test, and fermentative reactions on Triple sugar iron agar (US FDA 2023). Presumptive isolates were streaked onto a chromogenic medium (Hichrome vibrio agar, HiMedia, India) for further confirmation and the characteristic bluish-green colonies were transferred to Tryptic soy agar (HiMedia, India) supplemented with 3% NaCl and stored for further molecular identification.

2.3. Molecular confirmation of *V. parahaemolyticus* by PCR-based detection of species-specific genes

Genomic DNA extraction of the *V. parahaemolyticus* strains was carried out using the direct boiled cell lysate method (Letchumanan *et al.* 2015). PCR-based detection of species-specific *tlh* gene and *toxR* genes was performed using the primers and PCR reaction conditions described by Bej *et al.* (1999) and Kim *et al.* (1999) as given in Table 1.

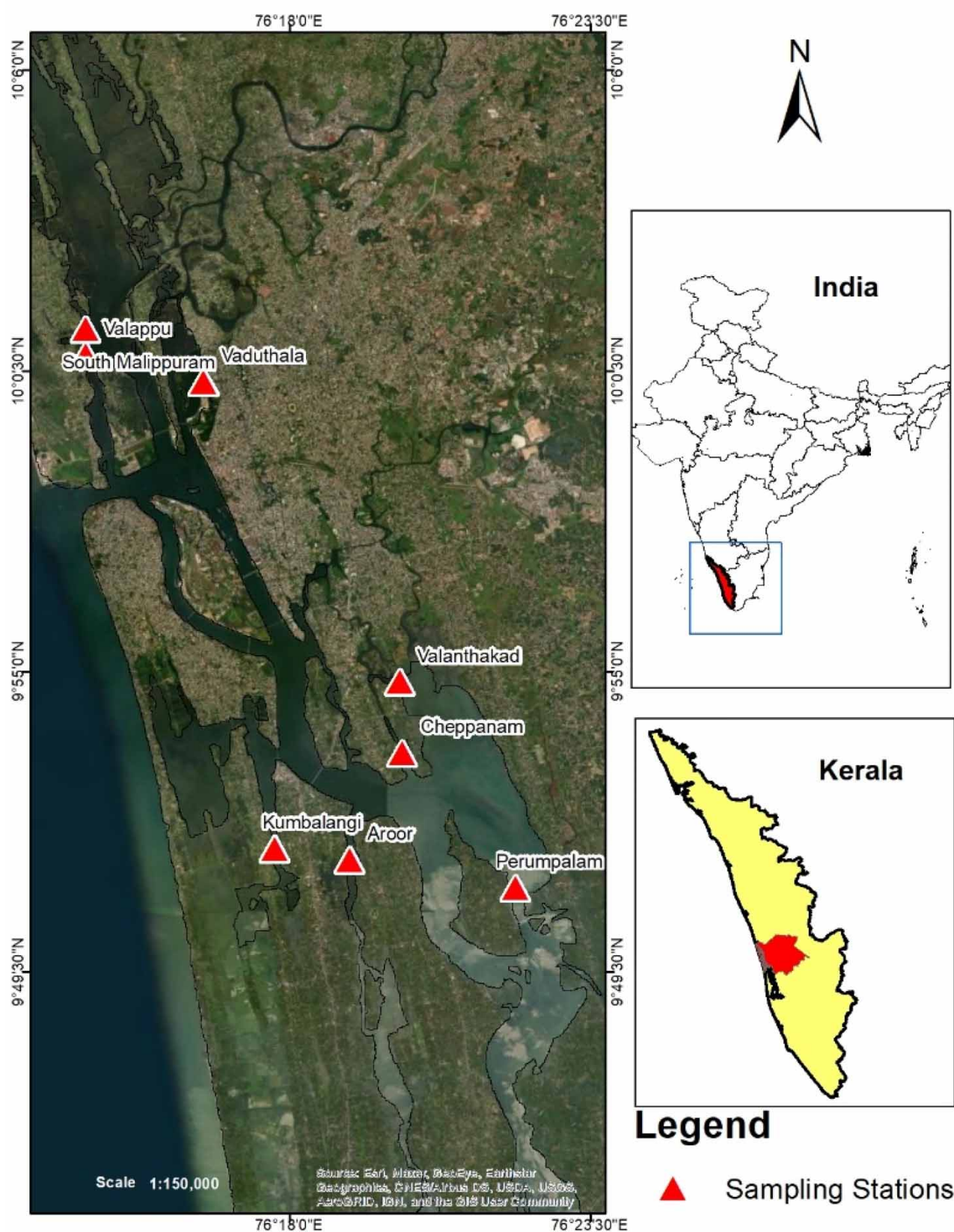


Figure 1 | Cochin estuary map showing the study area.

Table 1 | Detection of species-specific *tlh*, *toxR* genes, and virulence-associated *tdh* and *trh* genes

Organism	Type of PCR	Target gene (size, bp)	Primer sequence (5'-3')	PCR conditions	Reference
<i>V. parahaemolyticus</i>	Multiplex	<i>tlh</i> (450)	AAA GCG GAT TAT GCA GAA GCA CTG	94 °C, 3 min	Bej <i>et al.</i> (1999)
			GCT ACT TTC TAG CAT TTT CTC TGC	94 °C, 1 min	
		<i>tdh</i> (270)	GTAAAGGTCTCTGACTTTTGGAC	60 °C, 1 min	
	Simplex		TGGAATAGAACCCTTCATCTTCACC	72 °C, 2 min	Kim <i>et al.</i> (1999)
		<i>trh</i> (500)	TTGGCTTCGATATTTTCAGTATCT	72 °C, 3 min	
			CATAACAAACATATGCCCATTTCCG	30 cycles	
	Simplex	<i>toxR</i> (368)	GTCTTCTGACGCAATCGTTG	94 °C, 3 min	
			ATACGAGTGGTTGCTGTCATG	94 °C, 1 min	
				63 °C, 1.5 min	
				72 °C, 1.5 min	
				72 °C, 5 min	
				25 cycles	

2.4. Antibiotic susceptibility testing

The antibiotic sensitivity of the strains was determined by the agar disc diffusion method (Bauer *et al.* 1966) on Mueller-Hinton agar (MHA; HiMedia, India). Sixteen antibiotics (concentration (mcg) in parenthesis) were used: Amikacin (AK-30), Gentamicin (GEN-10), Streptomycin (S-10), Cefotaxime (CTX-30), Cefoxitin (CX-30), Cefpodoxime (CPD-10), Nitrofurantoin (NIT-100), Ampicillin (AMP-10), Amoxycylav (AMC-30), Chloramphenicol (C-30), Trimethoprim (TR-5), Nalidixic acid (NA-30), Ciprofloxacin (CIP-5), Tetracycline (TE-30), Doxycycline (DO-30), and Cotrimoxazole (COT-30).

Overnight cultures of the test organisms were swabbed on the surface of dried sterile MHA plates. After drying for 5 min, the antibiotic discs were aseptically placed over the seeded MHA plates, sufficiently separated from each other. After overnight incubation at 37 °C, the diameter of the inhibition zones was measured and interpreted according to the Clinical and Laboratory Standards Institute guidelines (CLSI 2023). Antimicrobial resistance (AMR) percentage against a single antibiotic was calculated using the formula a/b , where a represents the number of *V. parahaemolyticus* strains resistant to a single antibiotic, and b represents the total number of *V. parahaemolyticus* strains tested for susceptibility to the single antibiotic. The multiple antibiotic resistance (MAR) index of the individual test organism was calculated by the formula a/b , where a represents the number of antibiotics to which the isolate was resistant, and b represents the total number of antibiotics to which the isolate was exposed (Krumperman 1983). The strains with a MAR index greater than 0.2 are considered to have originated from high-risk sources of contamination where antibiotics are frequently used.

2.5. Multiplex PCR-based detection of virulence genes in *V. parahaemolyticus*

The presence of virulence genes, thermostable direct haemolysin (*tdh*) and thermostable haemolysin-related haemolysin (*trh*) were assessed by multiplex PCR using the primers and following the protocol described by Bej *et al.* (1999) (Table 1).

2.5.1. Gel documentation and image analysis

The PCR products were then electrophoresed on a 1.5% agarose gel (HiMedia, India) in 1× TBE Buffer (HiMedia, India) containing 0.5 µg/mL of ethidium bromide (HiMedia, India) and visualized by Gel Documentation system (GelDoc EZ Imager, BioRad, USA). The amplicon sizes were compared with a 100 bp DNA ladder.

2.6. Statistical analysis

Statistical analysis of the results was performed using SPSS software 20 (Statistical Package for Social Science). The chi-square test was applied to test the difference in the antibiotic resistance of bacterial strains from different sources. The significance level was set at 5%.

3. RESULTS

3.1. Isolation and molecular confirmation of *V. parahaemolyticus*

Isolation of *V. parahaemolyticus* was initially attempted using the conventional TCBS agar and biochemical methods. However, the methods proved highly inconclusive; hence, more selective Hichrome vibrio agar and PCR-based molecular

detection were employed for the detection of *V. parahaemolyticus* in the present study. Using the conventional approach, 317 presumptive *V. parahaemolyticus* colonies, (green colonies on TCBS agar) were isolated from shellfish, sediment, and harvesting water samples. Among the presumptive isolates, 62 (19.6%) formed typical blue-green colonies on Hichrome vibrio agar. All these 62 isolates exhibited PCR-based amplification of species-specific *tlh* and *toxR* genes, which confirmed them as *V. parahaemolyticus*.

3.2. Occurrence of *V. parahaemolyticus* in shellfish harvesting areas of the Cochin estuary

V. parahaemolyticus was isolated from all the eight harvesting areas under study. Out of the total 72 samples (shellfish $n = 24$, sediment $n = 24$, harvesting water $n = 24$) tested, 55.6% showed the presence of *V. parahaemolyticus*. Among the shellfish samples, 83.33% (20) showed the presence of *V. parahaemolyticus*, whereas 50% ($n = 12$) of sediment samples and 33.33% ($n = 8$) of water samples revealed the presence of *V. parahaemolyticus*. Out of the total 62 *V. parahaemolyticus* strains confirmed, 22 (35.5%) were isolated from shellfish, 26 (41.9%) from sediment, and 14 (22.5%) from water.

3.2.1. Detection of *V. parahaemolyticus* in shellfish harvesting areas of the Cochin estuary during various seasons

V. parahaemolyticus was isolated from shellfish (*Villorita cyprinoides*), sediment, and harvesting water samples of the Cochin estuary during various seasons as presented in Table 2. Maximum isolation of *V. parahaemolyticus* was obtained from shellfish samples compared with sediment and harvesting waters during all the three seasons.

3.3. Antibiotic resistance of *V. parahaemolyticus* isolated from shellfish harvesting areas of the Cochin estuary

Among the *V. parahaemolyticus* strains isolated 87% ($n = 54$) were MDR and exhibited 21 diverse antibiotic resistance patterns. The MAR index values ranged from 0 to 0.8 with an average of 0.44 ± 0.22 . The most repeated antibiotic resistance patterns were AMP, CX, and CPD. Maximum resistance was shown against ampicillin and cefpodoxime (93.5% each). Lower resistances were shown against tetracycline (9.7%) and chloramphenicol (6.5%). However, all the *V. parahaemolyticus* strains isolated were sensitive towards doxycycline.

3.3.1. Relative prevalence of antibiotic resistance among *V. parahaemolyticus* isolated from shellfish, sediment, and harvesting waters from shellfish harvesting areas

The percentage of antibiotic-resistant *V. parahaemolyticus* strains from the three sources (shellfish, sediment, and harvesting waters) against each antibiotic was calculated separately and the significance of which was statistically analysed using chi-square test. *V. parahaemolyticus* from all three sources showed significant differences in their antibiotic resistance (Figure 2). The strains from shellfish showed maximum relative resistance against three antibiotics ampicillin, cefoxitin, and cefpodoxime ($P < 0.05$), whereas those from sediment showed the highest resistance against four antibiotics, namely, amikacin, ampicillin, gentamicin, and streptomycin ($P < 0.05$). *V. parahaemolyticus* strains from harvesting waters showed maximum relative resistance against nine antibiotics, namely, amoxiclav, cefotaxime, chloramphenicol, ciprofloxacin, cotrimoxazole, nalidixic acid, nitrofurantoin, and trimethoprim ($P < 0.05$). Resistance to chloramphenicol was found only among *V. parahaemolyticus* from harvesting waters, while those from all the three sources were sensitive towards doxycycline. In general,

Table 2 | Occurrence of *V. parahaemolyticus* in shellfish harvesting areas of the Cochin estuary during various seasons

Stations	Pre-monsoon			Monsoon			Post-monsoon		
	Shellfish	Sediment	Water	Shellfish	Sediment	Water	Shellfish	Sediment	Water
1	–	–	+	+	+	+	–	–	+
2	+	–	–	–	+	–	+	–	–
3	+	–	+	–	–	+	+	–	–
4	+	–	–	+	–	–	+	–	–
5	+	+	–	+	+	–	+	–	–
6	+	+	–	+	+	+	+	+	+
7	+	–	–	+	+	–	+	+	–
8	+	+	+	+	+	–	+	+	–

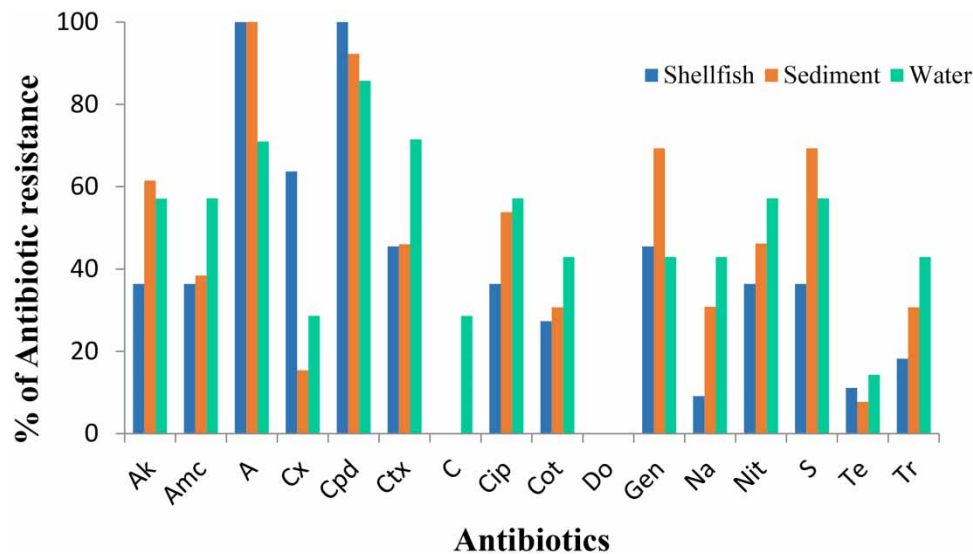


Figure 2 | Sample-wise comparison of antibiotic-resistant *V. parahaemolyticus* isolated from shellfish, sediment, and harvesting waters of the Cochin estuary.

V. parahaemolyticus strains isolated from harvesting waters tended to be resistant to more number of antibiotics than those isolated from sediment or shellfish ($P < 0.05$).

3.3.2. MAR index and antibiotic resistance patterns among *V. parahaemolyticus* from the Cochin estuary

Among the *V. parahaemolyticus* strains isolated from shellfish, 91% ($n = 20$) of them were MDR and exhibited eight different antibiotic resistance patterns (Figure 3). The MAR index values ranged from 0.13 to 0.8 with an average of 0.37 ± 0.22 . The most repeated antibiotic resistance patterns were AMP, CX, and CPD.

Among the strains from sediments 84.6% ($n = 22$) were MDR and 11 different resistance patterns were observed (Figure 3). The MAR indices ranged from 0.13 to 0.88 with an average of 0.37 ± 0.22 . The most repeated antibiotic resistance patterns were AMP, CPD, AK, AMP, CPD, CTX, CIP, COT, GEN, NA, NIT, S, and TR.

In harvesting waters, 85.7% ($n = 12$) of the total strains were MDR and five different antibiotic resistance patterns were observed (Figure 3). The MAR indices ranged from 0.13 to 0.88 with an average of 0.47 ± 0.33 . The most repeated antibiotic resistance pattern was found to be AK, AMC, AMP, CPD, CTX, CIP, GEN, NA, NIT, S, and TR.

3.4. Detection of virulence genes in *V. parahaemolyticus*

Virulence genes could be detected in 25.8% ($n = 16$) of the *V. parahaemolyticus* strains. *Trh* gene alone was found in 19.35% ($n = 12$) of strains, while 3.2% ($n = 2$) were positive for *tdh* gene alone. The coexistence of both *trh* and *tdh* genes was found in 3.2% ($n = 2$) of the total strains (Figure 4).

Sediment samples showed the maximum occurrence of the *trh* gene (eight strains), followed by shellfish samples (four strains). The *tdh* gene could be detected in two strains from shellfish. The coexistence of *tdh* and *trh* genes was detected in two strains from sediment samples. However, *V. parahaemolyticus* strains from harvesting waters lacked both virulence genes.

4. DISCUSSION

The presence of pathogenic and MDR *V. parahaemolyticus* strains in shellfish harvesting areas of the Cochin estuary indicates the possible safety hazard associated with the consumption of shellfish harvested from the area. Shellfish tend to accumulate *V. parahaemolyticus* in their tissues and it has been reported that their numbers sometimes even exceed the requirement for an infectious dose (Klein & Lovell 2017), which may result in a disease outbreak, if the strains are potentially virulent. *V. parahaemolyticus* was isolated from all eight shellfish harvesting areas selected for the study. Being a native of estuarine, marine, and coastal surroundings, its presence in shellfish growing areas has been widely reported (Letchumanan *et al.* 2014; Mannas *et al.* 2014).

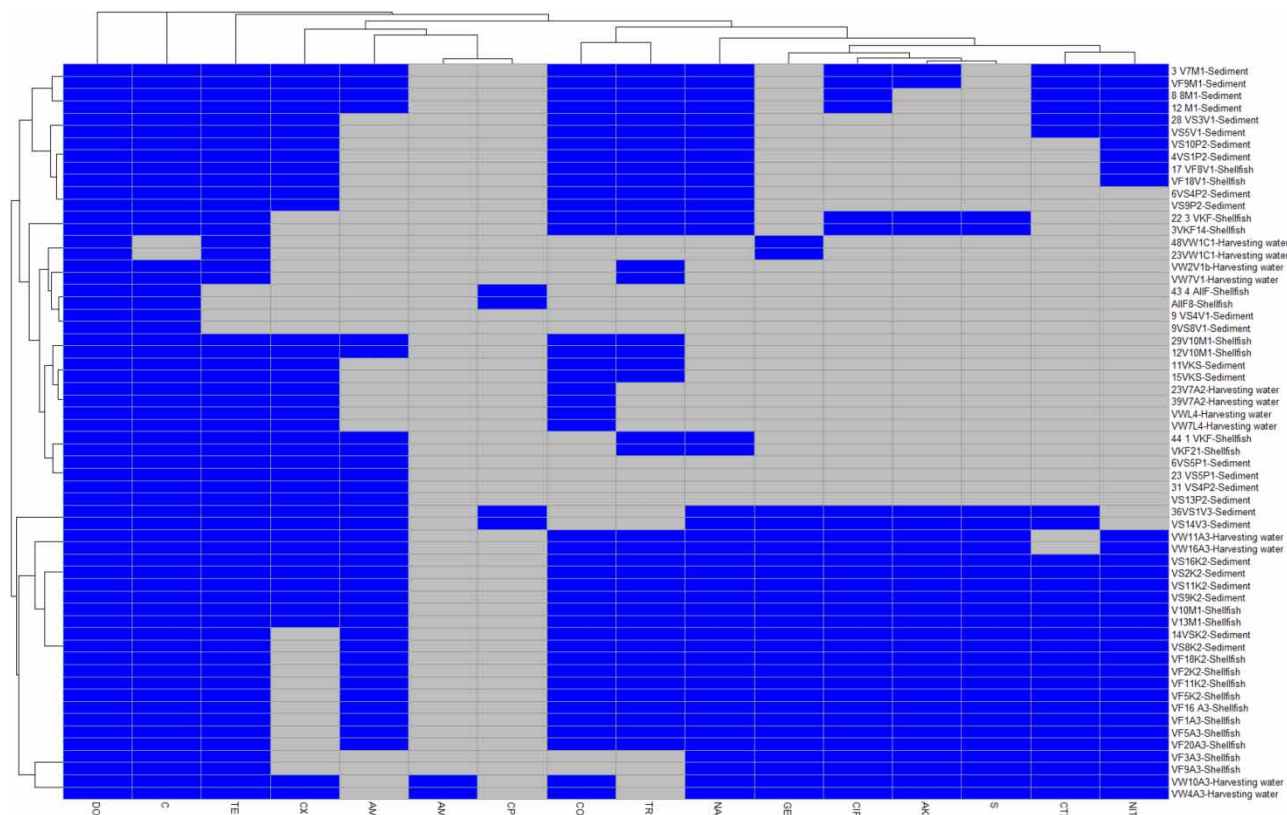


Figure 3 | Heat map of binary presence/absence data showing clustering of *V. parahaemolyticus* strains from shellfish, sediment, and harvesting waters based on antibiotic resistance pattern. Blue indicates resistant; while white indicates sensitive strains.

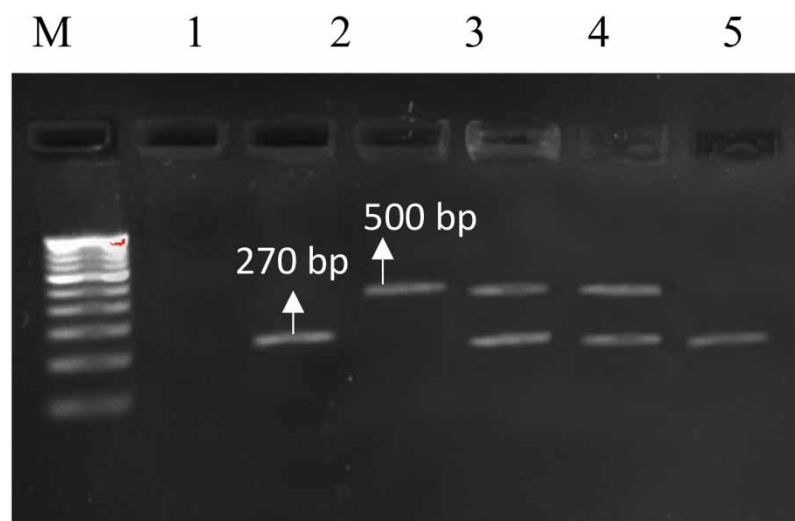


Figure 4 | Gel image showing amplified PCR products indicative of *tdh* (270 bp) and *trh* (500 bp) genes of *V. parahaemolyticus*. Lane M: 100 bp DNA marker; lane 1: negative control; lane 2: positive control for *tdh* gene; lane 3: positive control for *trh* gene; lanes 4 and 5: *tdh* and *trh* genes; lane 5: *tdh* gene.

Seasonal variation in the prevalence of *V. parahaemolyticus* has been widely reported (Johnson *et al.* 2010; Urquhart *et al.* 2015); often linking increased concentrations of the bacterium in water samples and subsequent isolations from shellfish to warmer temperatures (Schets *et al.* 2010). However, during our study, no seasonal association with the isolation of

V. parahaemolyticus was not observed as it could be isolated during all seasons. Seasonal influence probably may be more prominent in temperate regions, where favourable warmer temperatures are reached only during summer seasons, which thus results in enhanced bacterial multiplication and improved isolations during these warmer months (DePaola *et al.* 2000; Su & Liu 2007). Such temperature variations between seasons are not so prominent in tropical regions, where warmer water temperatures exist year-round. Supporting this fact, the temperatures recorded during the current study ranged from 28 to 33 °C (data not included) which included the pre-monsoon, monsoon, and post-monsoon seasons. Shellfish samples exhibited the maximum prevalence of *V. parahaemolyticus* (83.33%) followed by sediment (50%) and water samples (33.33%). A similar observation was made by Yu *et al.* (2013) in Taiwan, where increased prevalence was observed in oyster and clam samples followed by sediment samples, while water samples exhibited the least prevalence.

The extensive use and misuse of antibiotics have resulted in the emergence of antibiotic resistance among several human pathogens, reducing the possibilities for treatment of infections (Davies & Davies 2010). Antibiotic resistance has long been considered a clinical problem, however, recently there is increasing evidence for the role played by environments in the dissemination of antibiotic resistance (Martinez 2012). Eighty-seven percent of *V. parahaemolyticus* isolated from shellfish harvesting areas of the Cochin estuary in the study were found to be resistant to multiple antibiotics, which is very high compared with the findings of Manjusha *et al.* (2005), where 54% multiple drug resistance was reported from the same study area. *V. parahaemolyticus* isolated during the present study showed maximum resistance against ampicillin and cefpodoxime (93.5% each). This is in agreement with the findings of Silvester *et al.* (2015) and Sudha *et al.* (2014), where similar higher degrees of ampicillin resistance were reported in *V. parahaemolyticus* isolated from the same study area. In agreement with our findings, several previous studies have reported ampicillin resistance in *V. parahaemolyticus* from shellfish worldwide (Reboucas *et al.* 2011; Labella *et al.* 2013; Ottaviani *et al.* 2013; Siddique *et al.* 2021). Sudha *et al.* (2014) reported absolute sensitivity to antibiotics such as ciprofloxacin, nalidixic acid, streptomycin, and tetracycline among *V. parahaemolyticus* isolated from shellfish collected from markets of the study area. However, in our study, enhanced resistance against the aforesaid antibiotics was observed. The majority of the antibiotic classes such as penicillins, cephalosporins, and aminoglycosides against which maximum resistance was exhibited are not commonly used in aquaculture practices in India (Manjusha *et al.* 2005). Hence, it can be presumed that anthropogenic factors such as hospital effluents carrying these drugs might have been responsible for creating selection pressure for the emergence of drug resistance in vibrios from shellfish harvesting areas of the Cochin estuary. Interestingly, all the strains isolated from the study area exhibited maximum sensitivity towards tetracyclines and quinolones/fluoroquinolones which are commonly used in the aquaculture sector in India, which indicates either the aquaculture farms in the study area follow restricted antibiotic usage, or that the discharge from the aquaculture farms is well regulated. *V. parahaemolyticus* strains from harvesting waters exhibited maximum antibiotic resistance followed by those from sediment and shellfish. A previous study reported the presence of mobile genetic elements such as plasmids in *V. parahaemolyticus* from the study area (Silvester *et al.* 2015), which may facilitate horizontal transmission of AMR genes. It may thus serve as a permanent reservoir of antibiotic-resistant genes, resulting in AMR, cycling through the environment, food, and human sources.

Virulence genes *tdh* and *trh* could be detected in *V. parahaemolyticus* isolated from both shellfish and sediment samples during our study. Among the isolated strains, 22.5% were positive for *trh* genes, while only 6.4% were positive for *tdh* genes. Two strains from sediments showed the coexistence of both genes. In agreement with our results, two recent studies from retail seafood market samples from our study area reported a lesser *tdh* incidence of 2.8% (Silvester *et al.* 2015) and 6.8% (Narayanan *et al.* 2020), respectively. Concordant observations were made by several previous findings where low frequencies of *tdh* and *trh* genes were reported from most environmental *V. parahaemolyticus* strains (Ceccarelli *et al.* 2013; Siddique *et al.* 2021). Many researchers have the opinion that environmental strains have low frequencies of virulence genes and hence are not pathogenic to humans. In our study, the prevalence of *trh* genes (22.5%) was relatively higher than that of *tdh* genes (6.4%). This is in agreement with the findings of Nakaguchi (2013), where a higher comparative prevalence of the *trh* gene than the *tdh* gene was reported in molluscan shellfish from Malaysia and Indonesia. Baker-Austin *et al.* (2008) have also made similar observations in environmental strains from highly populated areas of South Carolina and Georgia coasts where a higher prevalence of *trh* gene was detected in *V. parahaemolyticus* strains from sediment samples, than those from shellfish samples in the present study. However, a few previous studies have reported higher frequencies of detection of virulence genes even from environmental strains (Klein & Lovell 2017).

Even though it is widely accepted that virulence genes are associated with pathogenic *V. parahaemolyticus*, nontoxigenic *V. parahaemolyticus* strains causing acute gastroenteritis have also been reported (Ottaviani *et al.* 2012). The absence of

virulence factors from some clinical strains (Ottaviani *et al.* 2012; Ronholm *et al.* 2016) as well as their occurrence in numerous environmental isolates that are apparently not acquired from infected humans (Gutierrez West *et al.* 2013), suggests the possibility that *tdh* and *trh* may serve some environmental roles unrelated to human disease (Gutierrez West *et al.* 2013), although the nature of these functions remain unknown. A study by Vongxay *et al.* (2008) suggested that haemolysins *tdh* and/or *trh* may not be necessarily the only virulence factors of pathogenic *V. parahaemolyticus* isolates. Hence, another possibility could be the presence of some other virulence factors such as type three secretion systems or some other factors which are yet to be identified.

5. CONCLUSION

In summary, our results confirm that the shellfish harvesting areas along the Cochin estuary can act as potential environmental reservoirs of pathogenic and MDR *V. parahaemolyticus*, which may contaminate the aquaculture products harvested from the study area. Maximum resistance was exhibited towards those antibiotic classes not used in aquaculture, which indicates anthropogenic involvement such as contamination from hospital effluents. In addition, they also have been found to harbour several virulence genes which could be horizontally transmitted, if located on mobile genetic elements. To the best of our knowledge, this is the first report of pathogenic *V. parahaemolyticus* from shellfish harvesting areas of the Indian subcontinent and demonstrates the presence and coexistence of *tdh*, *trh* virulence genes, as well as multiple drug resistance. The present study reveals the probable health hazards associated with the consumption of shellfish harvested from this estuary and emphasizes the necessity of regulatory interventions and continuous monitoring to detect the emergence of such pathogenic strains.

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DATA AVAILABILITY STATEMENT

All relevant data are included in the paper or its Supplementary Information.

CONFLICT OF INTEREST

The authors declare there is no conflict.

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